

Portfolio of Burcu Erarslanoglu

Machine Learning • Signal Processing • Medical Imaging

Email: burcuerarslanoglu@gmail.com

Phone: +31 6 33 09 11 71

GitHub: <https://github.com/burcuerarslanoglu>

Gaussian Process Regression on a Grid with Tensor Train Matrices

Sep 2024 - Nov 2024

Implemented Gaussian Process Regression for large grid-structured datasets using tensor train representations to exploit Kronecker structure in covariance matrices. Demonstrated significant reduction in computational complexity while preserving prediction accuracy.

Hand Gesture Classification Based on EMG Signals

Sep 2024 - Nov 2024

Developed a machine learning pipeline for hand gesture classification from EMG signals collected across multiple subjects. Extracted features in both time and frequency domains and evaluated KNN, SVM, and MLP classifiers using cross-validation. Optimized model performance through hyperparameter tuning and comparative analysis across classifiers.

Methods to Derive FC with fMRI Using BSS with Tensor Decomposition

Apr 2024 - Aug 2024

Reviewed blind source separation and tensor decomposition methods for extracting functional connectivity from fMRI data. Conducted comparative analyses of these methods and assessed their strengths and limitations for high-dimensional neuroimaging applications.

Linear Discriminant Analysis for Recognising Differences

Apr 2024 - Jun 2024

Applied a PCA-LDA framework to MRI brain scans to classify Alzheimer's disease through dimensionality reduction. Evaluated classification accuracy across different numbers of principal components using k-fold cross-validation, demonstrating potential for early diagnosis of neurodegenerative diseases.

Array Parameter Estimation and Speech Enhancement for Array Processing

Apr 2024 - Jun 2024

Studied methods used in the estimation of directions and frequencies in an array and channel equalization. Compared the performances of EVD and GEVD on the three different beamformers by using SNR and STOI metrics.

Wavefield Imaging on 2D Linearized Inverse Scattering Problems

Apr 2024 - Jun 2024

Solved a 2D linearized inverse scattering problem to reconstruct an image of an object from wavefield measurements using contrast function estimation.

Boolean Compressed Sensing

Feb 2024 - May 2024

Implemented and compared several sparse recovery algorithms for Boolean compressed sensing in the context of group testing.

Compressed Sensing and Machine Learning Techniques in MRI

Feb 2024 - May 2024

Reviewed compressed sensing and machine learning approaches for accelerated MRI, covering techniques and applications across both domains. Implemented a compressed sensing + DnCNN pipeline in MATLAB for brain MRI reconstruction. Showed that ML-assisted undersampling can recover SNR comparable to fully sampled acquisitions while reducing scan time.

Towards Reconstructing the Brain with Graph Signal Processing

Feb 2024 - Apr 2024

Conducted a literature review on graph signal processing (GSP) for modelling structural and functional brain connectivity. Surveyed approaches combining GSP with deep learning for neuroimaging analysis and connectivity matrix generation, and assessed their potential for mental disorder diagnosis.

Compressed Sensing for Applied Convex Optimisation

Nov 2023 - Jan 2024

Formulated signal reconstruction as a convex optimization problem and implemented a compressed sensing-based recovery algorithm. Reconstructed signals using both off-the-shelf solvers and a custom low-complexity projected subgradient descent algorithm, and compared their performance.

Accelerometer Calibration for Estimation and Detection

Nov 2023 - Jan 2024

Various estimators like the Maximum Likelihood Estimator (MLE) and performance bounds like the Cramer-Rao

Lower Bound (CRLB) were applied for the challenge of calibrating the accelerometer.

Fusing Noisy and Corrupted MRI Images

Sep 2023 - Nov 2023

Developed a pipeline to detect and restore outliers in eye MRI scans caused by patient motion and noise, using optimal filtering and spectral estimation techniques. Implemented and compared multiple filters, evaluating image quality improvements quantitatively using SNR.

Online Music Platform Database Application

Feb 2023 - June 2023

Designed and implemented a database-backed online music platform supporting multiple user roles. The system enabled operations such as track browsing/filtering, album/playlist management, user history tracking, and social interactions including following users and liking playlists. The project involved designing relational database schemas and implementing queries for managing music metadata, user activity, and artist collaborations.

Industrial Design Project: Detection of Forest Fires with WSNs

Sep 2022 - July 2023

This project is carried out with METEKSAN Defence on the scope of EEE 493/494 which is a capstone design course. By considering the placement of sensors, power efficiency, communication protocols and terrain propagation effects; algorithms will be proposed and tested with MATLAB simulations for the detection of forest fires with WSNs.

Medical Imaging Project: Brain MRI Simulator

Dec 2022 - Jan 2023

Built an MRI simulator from scratch to cover the full acquisition pipeline: slice selection, k-space sampling, and image reconstruction. Studied the main contrast mechanisms of MRI, namely T_1 , T_2 , PD weighting, and flip angle, alongside Gaussian and spike noise effects. Investigated the effect of each contrast mechanism and noise type on simulated brain MR images. [Video Link](#).

CMOS VLSI Design Project: SRAM Design

May 2022 - Jun 2022

In this project, an 8x4 memory array and an 8x4 SRAM array was designed and combined in Cadence Virtuoso. Testbench results were obtained and power optimization was done. [Video Link](#).

Electronic Circuit Project: Design of D and T Flip-Flop

Sep 2021 - Dec 2021

The design of D and T Flip-Flop was done by using BJTs. The proposed design was simulated on LTspice and tested on a breadboard. The proposed design was implemented on a PCB which was designed using DipTrace.

Microprocessors Project: Smart Car

Feb 2021 - May 2021

In this project, FRDM-KL25Z board was used and the coding was done in C++ through the Mbed Compiler. The main components of the project were TCS3200 color sensor, HC-SR04 ultrasonic proximity sensor, L298N dual full-bridge driver, 4x4 keypad and 8x8 dot matrix board. [Video Link](#).

Digital Circuit Design Project: Weather Station

Sep 2020 - Dec 2020

In this project, BASYS3 board was used and the coding was done in VHDL. The main components of the project were DHT22 sensor, LPS25HB sensor, 16x2 LCD display and a VGA monitor. [Video Link](#).